GITAM Deemed to be University

Electrical and Electronics Engineering | Course Code: EE304

Power Electronics & Converters - Laboratory Manual & Practical Guide

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Syllabus Focus: EE304 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Power Semiconductor Switches

" Syllabus Scope: Thyristors SCR, Triac, Power MOSFET, IGBT, Switching characteristics, Safe Operating Ar

Unit 2: Phase-Controlled Converters

" Syllabus Scope: Single-phase and three-phase half-wave and full-wave controlled rectifiers with R, RL,

EE304: Power Electronics & Converters - Advanced Topics & Assessment

Academic Track: Electrical and Electronics Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: DC-DC Switched Mode Regulators

" Buck converter, Boost converter, Buck-Boost topology, Continuous and discontinuous conduction boundary modes

Unit 4: Inverters & Modulation Techniques

" Voltage Source Inverters VSI, Single-phase full-bridge, Three-phase 120 and 180 conduction modes, Sinusoidal

Unit 5: AC Voltage Controllers & Cycloconverters

" Single-phase bidirectional controllers with resistive and inductive loads, Sequence control, Cycloconverter

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: V-I Characteristics of SCR, MOSFET, and IGBT under Varying Gate Trigger Voltages

Lab 2: Single-Phase Fully-Controlled Thyristor Bridge Rectifier with R and RL Load Testing

Lab 3: Design and Closed-Loop Output Voltage Regulation of a DC-DC Buck Converter

Lab 4: Speed Control of a DC Motor using a Microcontroller-Driven H-Bridge PWM Driver

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Ned Mohan, Tore M. Undeland - Power Electronics: Converters, Applications, and Design Wiley 3rd Ed

* Muhammad H. Rashid - Power Electronics: Devices, Circuits, and Applications Pearson 4th Ed

Authorized NoteVault Educational Document " Verified for Registered University Students